BCUN: Bayesian Fully Convolutional Neural Network for Hyperspectral Spectral Unmixing

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Abstract—**Spectral unmixing (SU) plays a fundamental role in hyperspectral image (HSI) processing. Effective SU relies on the accurate and efficient characterization of the noise effect, the endmembers, and the spatial correlation effect in abundances, as well as efficient optimization techniques to estimate these effects. To address these issues, this article presents a Bayesian fully convolutional hyperspectral unmixing network (BCUN) with the following key characteristics. First, a fully convolutional neural network (FCNN)-based deep image prior (DIP) is designed for enhanced characterization and estimation of the spatial context information in abundance maps, leading to more efficient and accurate abundance modeling than the traditional nonnegative least squares (NNLS) approaches. Second, a multivariate Gaussian distribution with an anisotropic covariance matrix is designed to characterize the conditional distribution of the spectral observations, leading to a novel Mahalanobis distance-based loss for FCNN training that is better capable of addressing the noise heterogeneous effect in HSI than the Euclidean distance-based mean squared error (MSE) loss in traditional deep neural networks. Third, the designed conditional distribution of spectral observations also enables the incorporation of the spectral mixture model (SMM) into the FCNN training process for effectively leveraging the knowledge in the forward spectral model. Fourth, the endmembers are modeled and estimated by a "purified means" approach that is capable of better characterizing endmembers. Finally, the above key components are coherently integrated into a Bayesian framework, and the resulting maximum *a posteriori* (MAP) problem is solved by a designed expectation–maximization (EM) algorithm. Experimental results on both simulated and real HSIs demonstrate that the proposed BCUN approach outperforms the other classical and state-of-the-art methods on both endmember estimation and abundance estimation.**

Index Terms—**Bayesian framework, deep image prior (DIP), fully convolutional neural network (FCNN), heterogeneous noise, hyperspectral image (HSI), spatial correlation, spectral unmixing (SU).**

I. INTRODUCTION

HYPERSPECTRAL imaging is a rapidly growing remote sensing technique, which records the electromagnetic radiation from the Earth's surface via hundreds of narrow and contiguous spectral bands [1]. With rich spectral information, hyperspectral images (HSIs) are critical to a wide variety of applications, including ground target classification [2]–[4], subpixel mapping [5], agricultural management [6], and environmental monitoring [7], [8]. However, due to the limitation in the spatial resolution of HSI and the complexity of ground targets, spectral pixels in HSI usually contain the spectral contribution of multiple pure materials, leading to mixed pixels in HSI. Spectral unmixing (SU) aims to decompose the mixed pixels to derive both the spectral signatures of constituent components (i.e., endmembers) and their corresponding fractional proportions (i.e., abundances) from the mixed pixel in HSI [9]. SU is a challenging ill-posed inverse problem [9]. An effective SU approach relies on accurately modeling of the noise effect, the endmembers, the spatial correlation effect of abundance, and an effective optimization approach for estimating these effects. However, most existing SU methods ignore or fail to fully explore these key factors. First, HSI contains not only rich spectral information but also abundant spatial information. The spatial context information in HSI is crucial for regulating and estimating endmembers and abundances in HSI [10]. Nevertheless, most existing SU methods ignore this spatial correlation effect by treating the abundance of each pixel as independently distributed variables using nonnegative least squares (NNLS) [11], sparse unmixing by variable splitting and augmented lagrangian (Sunsal) [12], or fully connected layers [13], [14] for abundance estimation. There is still a lack of advanced modeling and estimation approaches for addressing large-scale spatial correlation effects that commonly exist in HSI. Compared with the traditional patch-based convolutional neural network (CNN), fully CNN (FCNN) can better capture the large-scale spatial correlation effect in HSI [15], [16]. FCNN approaches have been widely used in many tasks, e.g., semantic segmentation [17], [18], super-resolution [19], and image denoising [20]. However, they

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are rarely adapted to HSI for SU. Although FCNN was used to efficiently map random noise to abundances in a supervised manner [21], it was less efficient in leveraging the forward spectral mixture model (SMM) for efficiently estimating both abundance and endmembers in an integrated manner. Therefore, how to integrate FCNN into SU for better modeling and estimation of spatial information in HSI is an important research issue.

Second, HSI is contaminated by noise in the data acquisition process. Due to the different spectral absorption properties of different spectral channels and the common existence of "junk bands," the noise level of different HSI channels tends to vary dramatically, causing different noise variances in different HSI channels [22]. Since SU is very sensitive to noise, the success of SU algorithms depends on their effectiveness in accurately accounting for and resisting the noise effect [9]. However, most of the existing methods assume that different bands in HSI contain the same degree of noise by using the mean squared error (MSE) as the reconstruction loss, leading to the undesirable preservation of noise in some bands but erasing of signal in some other bands. Consequently, the design of the model that is able to accommodate the noise variance heterogeneous effect in HSI is an important issue for SU.

Third, meaningful SU relies on effective constraints imposed on endmembers. However, existing endmember extraction methods either rely on the minimum volume simplex constraint (e.g., vertex component analysis (VCA) [23] and pixel purity index (PPI) [24]) that cannot deal with highly mixed pixels or on prior distribution constraints in the Bayesian framework that is computationally intractable [9]. "Purified means" approach [10], [25], [26] provides simple and effective constraints for deriving endmembers that are achieved by treating the endmember as the mean vector of the "purified" pixels. Adopting this efficient constraint on endmembers boosts the accuracy and effectiveness of the estimation of endmembers and abundances.

Fourth, given the above mentioned three key factors for SU, it is critical to design a coherent framework with effective modeling capacity and an efficient optimization approach, which is capable of capturing the large-scale spatial correlation effect in abundances, the variance heterogeneity effect of noise, and the proper constraint on endmembers. To solve ill-posed problems where endmembers and abundances are unknown underlying quantities, the SU process can be generalized as a maximum *a posteriori* (MAP) problem in a Bayesian framework and optimized by an expectation–maximization (EM) algorithm. The iterative method allows the estimated abundances and endmembers to be updated in an adaptive manner.

Therefore, in this article, following the linear SMM (LSMM) that is commonly used to describe the HSI generation process [27], we propose a Bayesian fully convolutional hyperspectral unmixing network (BCUN) that integrates FCNN and LSMM in a Bayesian framework for HSI unmixing.

In the proposed approach, the data likelihood is designed based on the LSMM, the conditional probability of the endmembers is derived from the "purified means" approach [10], [25], [26], and the prior of abundance is implemented by the deep image prior (DIP) [28]. The key contributions of this article are summarized as follows.

1) A skip-connection FCNN is designed for abundance estimation, where DIP is used to model the spatial correlation of the abundance field. Compared to NNLS or a fully connected network, the FCNN is able to efficiently and accurately estimate abundances by leveraging GPUs and the large-scale spatial correlation of HSI.
2) The noise is modeled as a multivariate Gaussian distribution to account for the noise variance heterogeneity in HSI. As a result, the loss function of BCUN is designed based on the M-distance rather than MSE loss. The designed conditional distribution of spectral observations also enables the incorporation of the SMM into the FCNN training process for effectively leveraging the knowledge in the forward spectral model.
3) The endmember is modeled and estimated by a "purified means" approach, which can be seamlessly integrated into the Bayesian framework by a designed conditional distribution of the endmembers given the abundance.
4) The above key components are coherently integrated into a Bayesian framework, and the resulting MAP problem is solved by a designed EM algorithm.

Experiments on both simulated and real HSI demonstrate that, compared with traditional and state-of-the-art approaches, the proposed BCUN can extract endmembers and map abundances more accurately by exploring the spatial correlation effect and noise heterogeneity effect in HSI. The remainder of the article is organized as follows. Section II formulates the unmixing problem in a Bayesian framework. The design of the network and its rationale are detailed in Section III. Section IV introduces the optimization scheme of BCUN. Section V conducts experiments on both simulated and real HSI.

II. PROBLEM FORMULATION

Assuming that there are P spectral bands and N pixels in an HSI, we denote the observed reflectance of the pixel at site i by $\boldsymbol{x}_i$, which is $P \times 1$ vector. Then, the HSI can be expressed as $\boldsymbol{X} = \{\boldsymbol{x}_i | i = 1, 2, \ldots, N\}$. The HSI is assumed to contain K endmembers. According to the LSMM, the observed HSI $\boldsymbol{X} \in \mathbb{R}^{P\times N}$ is represented by the product of the endmember matrix $\boldsymbol{A} \in \mathbb{R}^{P\times K}$ and the abundance matrix $\boldsymbol{S} \in \mathbb{R}^{K\times N}$, plus some Gaussian noise $\boldsymbol{N} \in \mathbb{R}^{P\times N}$, as follows:

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S} + \boldsymbol{N} \tag{1}$$

where $\boldsymbol{S} = \{\boldsymbol{s}_i | i = 1, \ldots, N\}$ and $\boldsymbol{A} = \{\boldsymbol{a}_k | k = 1, \ldots, K\}$. $\boldsymbol{s}_i$ is a K-dimensional vector, and $\boldsymbol{a}_k$ is a P-dimenstional vector. Then, the pixel $\boldsymbol{x}_i$ can be formulated as a linear combination of the endmembers $\boldsymbol{A}$ weighted by their associated abundances $\boldsymbol{s}_i$ plus noise $\boldsymbol{n}_i$

$$\boldsymbol{x}_i = \sum_{k=1}^{K} \boldsymbol{a}_k s_i^k + \boldsymbol{n}_i, \quad \sum_k s_i^k = 1 \quad \forall s_i^k > 0. \tag{2}$$

The noise distribution is assumed to satisfy a Gaussian model as follows:

$$p(\boldsymbol{n}_i) = \frac{1}{\sqrt{(2\pi)^P |\boldsymbol{\Lambda}|}} \exp\left(-\frac{1}{2}\boldsymbol{n}_i{}^T \boldsymbol{\Lambda}^{-1} \boldsymbol{n}_i\right) \tag{3}$$

where $\boldsymbol{\Lambda}$ is commonly expressed by the most existing methods as a diagonal matrix

$$\boldsymbol{\Lambda} = \begin{bmatrix} \sigma^2 & & & \\ & \sigma^2 & & \\ & & \ddots & \\ & & & \sigma^2 \end{bmatrix}$$

in which σ^2 is the noise variance of the each band.

The SU of HSI aims to infer the endmembers $\boldsymbol{A}$ and the abundance $\boldsymbol{S}$ based on the spectrum observation $\boldsymbol{X}$, which, in a Bayesian framework, can be achieved by maximizing the posterior distribution $p(\boldsymbol{\theta}|\boldsymbol{X})$, i.e.,

$$p(\boldsymbol{\theta}|\boldsymbol{X}) \propto p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta}) \tag{4}$$

where

$$\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{S}\}. \tag{5}$$

According to Bayes' theorem, the posterior distribution can be rewritten as

$$p(\boldsymbol{A}, \boldsymbol{S}|\boldsymbol{X}) \propto p(\boldsymbol{X}|\boldsymbol{A}, \boldsymbol{S})p(\boldsymbol{A}|\boldsymbol{S})p(\boldsymbol{S}). \tag{6}$$

Given the LSMM describing the generative model of HSI in 1 and the posterior distribution in 6, several key factors for effective SU are identified as follows.

1) First, the accurate modeling of the abundance prior $p(\boldsymbol{S})$ is critical for regulating and estimating the abundance $\boldsymbol{S}$.
2) Second, properly characterizing the noise $\boldsymbol{N}$ in HSI facilitates the accurate estimation of $\boldsymbol{A}$ and $\boldsymbol{S}$.
3) Third, meaningful constraints on endmembers $\boldsymbol{A}$ that guide and regulate the endmember estimation process.
4) Fourth, an efficient optimization scheme for solving the Bayesian inverse problem is critical.

In this article, $p(\boldsymbol{S})$ is achieved by Gaussian distribution where the spatial correlation effect in $\boldsymbol{S}$ is modeled by the DIP of FCNN, as detailed in Section III-A. Under the assumption that noise $\boldsymbol{n_i}$ has heterogeneous band-dependent noise variance, the data likelihood $p(\boldsymbol{X}|\boldsymbol{A}, \boldsymbol{S})$ is modeled by the anisotropic multivariate Gaussian distribution, as detailed in Section III-B. $p(\boldsymbol{A}|\boldsymbol{S})$ is achieved by Gaussian distribution where the expectation of $\boldsymbol{A}$ is derived by the purified means, as detailed in Section III-C. An efficient EM algorithm-based optimization scheme is designed and implemented in Section IV for solving the new Bayesian inverse problem.

III. BCUN: BAYESIAN FULLY CONVOLUTIONAL HYPERSPECTRAL UNMIXING NETWORK

A. Prior of Abundances

There are three key requirements on the abundance $\boldsymbol{S}$ when designing the abundance prior $p(\boldsymbol{S})$.

1) First, the large-scale spatial correlation effect in abundance should be fully exploited.
2) Second, abundances should be subject to the nonnegative and sum-to-one constraints, i.e., $\sum_k s_i^k = 1$ and $\forall s_i^k > 0$.
3) Third, abundances prior should allow efficient optimization.

Here, the prior of $\boldsymbol{S}$ is expressed as [29]–[31]

$$p(\boldsymbol{S}) = \frac{1}{z}\exp\left(-||\boldsymbol{S} - E(\boldsymbol{S})||^2\right) \tag{7}$$

where $E(\boldsymbol{S})$ is the expectation of $\boldsymbol{S}$, which is implemented as an FCNN. Comparing with patch-based CNN, FCNN has a wide field of view of the input image and enables better modeling of the spatial correlation effect in HSI.

The prior spatial information of abundances $\boldsymbol{S}$ can be captured by an FCNN structure [28], which has a wide field of view of the input image, and can be optimized efficiently on GPUs. Using $f(\cdot)$ to represent the FCNN forward propagation, the expected $\boldsymbol{S}$ is written as

$$E(\boldsymbol{S}) = f(\boldsymbol{Z}, \boldsymbol{\beta}) \tag{8}$$

where $\boldsymbol{Z}$ is the input random noise and $\boldsymbol{\beta}$ is the set of model parameters, including all convolution kernels and biases. The nonnegative and the sum-to-one constraint can be achieved using that "softmax" activation approach that is expressed as

$$\text{Softmax}(I) = \frac{e^{I_i}}{\sum_j^K e^{I_j}}. \tag{9}$$

In this work, we use a U-Net type "hourglass" architecture with skip connection [28] to model a mapping $f(\cdot)$ from the input variable $\boldsymbol{Z}$ to abundance maps $\boldsymbol{S}$.

As shown in Fig. 1, the FCNN network is a "hourglass" architecture with encoder and decoder parts, as well as the skip connection. The constitutional unit of the encoder part involves mainly two convolution operations (the yellow and blue blocks in Fig. 1), which can be formulated as:

$$\text{En}(I) = \text{LR}\left(\text{BN}\left(W_e^2 \circledast \left(\text{LR}\left(\text{BN}\left(W_e^1 \circledast I + B_e^1\right)\right)\right) + B_e^2\right)\right) \tag{10}$$

where I represents the input data, LR represents the "leaky ReLU" activation function, and BN denotes the batch normalization. $\circledast$ is the 2-D convolution operation. W and B are convolution kernels and the bias separately.

The skip connection operation mainly contains a convolution operation where the kernel size is illustrated in Fig. 1. The entire operation of skip connection is given as follows:

$$\text{Skip}(I) = \text{LR}(\text{BN}(W_s \circledast I + B_s)). \tag{11}$$

The unit operation of decoder (the red and purple blocks in Fig. 1) is expressed as

$$\text{De}(I) = \text{LR}\left(\text{BN}\left(W_d^2 \circledast \left(\text{LR}\left(\text{BN}\left(W_d^1 \circledast \text{BN}(I) + B_d^1\right)\right)\right) + B_d^2\right)\right). \tag{12}$$

The encoder part of FCNN decreases the size of the input via convolutions, and the decoder part recovers the image size by the decoder unit expressed in 12 and the upsampling operation. By connecting the corresponding parts that have the same scale of the encoder and the decoder, the spatial information in a certain scale is able to be well preserved. Thus, the resulting layer in a certain scale can be obtained by

$$\text{Scale}(I) = \text{De}(\text{up}(\text{En}(I)) \oplus \text{skip}(I)) \tag{13}$$

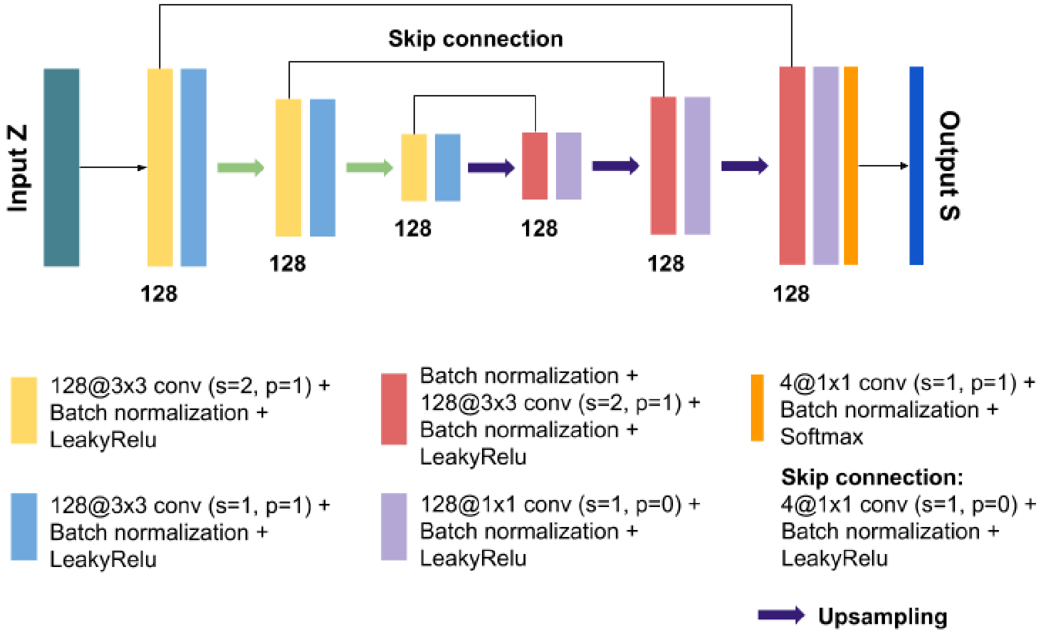

Skip connection
Input Z
128
128
128
128
128
128
Output S
128@3x3 conv (s=2, p=1) + Batch normalization + LeakyRelu
128@3x3 conv (s=1, p=1) + Batch normalization + LeakyRelu
Batch normalization + 128@3x3 conv (s=2, p=1) + Batch normalization + LeakyRelu
128@1x1 conv (s=1, p=0) + Batch normalization + LeakyRelu
4@1x1 conv (s=1, p=1) + Batch normalization + Softmax
Skip connection: 4@1x1 conv (s=1, p=0) + Batch normalization + LeakyRelu
Upsampling

Fig. 1. FCNN structure in the BCUN framework.

where $\oplus$ is the matrix concatenation operation. To preserve and utilize the spatial information of different image scale, the number of scale L is usually set as a number more than 1. Then, the main part of the FCNN is expressed as

$$\mathrm{SCALE}_L(I) = \mathrm{Scale}_L, \ldots, (\mathrm{Scale}_2(\mathrm{Scale}_1(I))). \tag{14}$$

In Fig. 1, $L = 3$, and the last layer is a convolution layer with the kernel size of 1×1 (the orange block in Fig. 1). The entire operation of the output layer is given as follows:

$$\mathrm{Out}(I) = \mathrm{BN}\left(W^o \circledast I + B^o\right). \tag{15}$$

By applying the "softmax" activation to the output layer, the whole FCNN is formulated by

$$f(I) = \mathrm{Softmax}(\mathrm{Out}(\mathrm{SCALE}_L(I))). \tag{16}$$

B. Data Likelihood

The formulation of data likelihood $p(\boldsymbol{X}|\boldsymbol{A}, \boldsymbol{S})$ relies on the probabilistic distribution of noise $\boldsymbol{N}$. The noise distribution is assumed to satisfy a Gaussian model as 3. in which $\boldsymbol{\Lambda}$ is the covariance matrix of noise. Traditionally, most methods assume hogmogeneous noise variance across different HSI bands and treat $\boldsymbol{\Lambda}$ as an isotropic diagonal matrix with the same diagonal elements, leading to the Euclidean distance-based loss. Instead, in this article, considering the variance heterogeneity of noise, $\boldsymbol{\Lambda}$ is expressed as follows:

$$\boldsymbol{\Lambda} = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_P^2 \end{bmatrix}$$

in which σ_p^2 is the noise variance of the pth band. This matrix allows different channels to have different noise variances, and thereby, it can better accommodate the noise variance heterogeneity issue for enhanced noise characterization and SU.

According to the LSMM in (2), the data likelihood is expressed as

$$p(\boldsymbol{X}|\boldsymbol{A}, \boldsymbol{S}) = \frac{1}{\sqrt{(2\pi)^P|\boldsymbol{\Lambda}|}}\exp\left(-\frac{1}{2}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S})^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S})\right) \tag{17}$$

where $(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S})^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S})$ is the reconstruction error that is essentially the square of the M-distance $Dist_M(\boldsymbol{X}, \hat{\boldsymbol{A}}\hat{\boldsymbol{S}})$ between the original HSI $\boldsymbol{X}$ and the reconstructed HSI $\hat{\boldsymbol{A}}\hat{\boldsymbol{S}}$.

C. Conditional Distribution of Endmembers Given Abundance

We choose a conditional prior distribution for the endmembers with the form of a Gaussian distribution [32], which encourages the small distance between the estimated endmember and its expectation

$$p\left(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right) = \frac{1}{z}\exp\left(-\left|\left|\boldsymbol{a}_k - E\left(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right)\right|\right|^2\right). \tag{18}$$

Then, the joint density $p(\boldsymbol{A}|\boldsymbol{S})$ is given as follows:

$$\begin{aligned} p(\boldsymbol{A}|\boldsymbol{S}) &= \prod_{j=1}^{K} p\left(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right) \\ &= \prod_{j=1}^{K} \frac{1}{z}\exp\left(-\left|\left|\boldsymbol{a}_k - E\left(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right)\right|\right|^2\right). \end{aligned} \tag{19}$$

The above implementation is based on a conditional independence assumption of endmember $\boldsymbol{a}_k$ given the rest of the endmembers $\boldsymbol{a}_{j\neq k}$.

IV. MODEL OPTIMIZATION

A. MAP Estimation

The unmixing problem in (4) can be solved by the MAP approach, where the endmembers $\boldsymbol{A}$ and the abundance $\boldsymbol{S}$ are estimated by maximizing the posterior distribution of $\{\boldsymbol{A}, \boldsymbol{S}\}$ given the observed HSI $\boldsymbol{X}$, i.e.,

$$\left\{\hat{\boldsymbol{A}}, \hat{\boldsymbol{S}}\right\} = \arg\max_{\boldsymbol{A},\boldsymbol{S}}\{p(\boldsymbol{A}, \boldsymbol{S}|\boldsymbol{X})\}. \tag{20}$$

Maximizing $p(\boldsymbol{A}, \boldsymbol{S}|\boldsymbol{X})$ is equivalent to minimizing its negative logarithm likelihood, i.e.,

$$\left\{\hat{\boldsymbol{A}}, \hat{\boldsymbol{S}}\right\} = \arg\min_{\boldsymbol{A},\boldsymbol{S}}\{-\log p(\boldsymbol{A}, \boldsymbol{S}|\boldsymbol{X})\}. \tag{21}$$

Then, the objective function can be written as

$$\begin{aligned} J_{A,S} &= \arg\min_{\boldsymbol{A},\boldsymbol{S}}\{-\log p(\boldsymbol{A}, \boldsymbol{S}|\boldsymbol{X})\} \\ &\propto \arg\min_{\boldsymbol{A},\boldsymbol{S}}\{-\log p(\boldsymbol{X}|\boldsymbol{A}, \boldsymbol{S}) - \log p(\boldsymbol{S}) - \log p(\boldsymbol{A}|\boldsymbol{S})\}. \end{aligned} \tag{22}$$

Considering (7), (17), and (19), the objective function in (22) can be reformulated as

$$\begin{aligned} J_{A,S} = \arg\min_{\boldsymbol{A},\boldsymbol{S}}\Bigg\{&\left((\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S})^T \boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S})\right) \\ &+ ||\boldsymbol{S} - E(\boldsymbol{S})||^2 \\ &+ \alpha\sum_{k=1}^{K}\left|\left|\boldsymbol{a}_k - E\left(\boldsymbol{a_k}|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right)\right|\right|^2\Bigg\}. \end{aligned} \tag{23}$$

In E-step of the EM algorithm, $\boldsymbol{S}$ is estimated by $E(\boldsymbol{S})$. Thus, we replace $\boldsymbol{S}$ with $E(\boldsymbol{S})$ to simplify the objective function. As a result, the objective function can be reformulated as [28]

$$\begin{aligned} J_{A,S} = \arg\min_{\boldsymbol{A},\boldsymbol{S}}\Bigg\{&\left((\boldsymbol{X} - \boldsymbol{A}E(\boldsymbol{S}))^T \boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}E(\boldsymbol{S}))\right) \\ &+\alpha\sum_{k=1}^{K}\left|\left|\boldsymbol{a}_k - E\left(\boldsymbol{a_k}|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right)\right|\right|^2\Bigg\} \end{aligned} \tag{24}$$

where α is a weighting parameter. This objective function has following characteristics.

1) The EM algorithm is used to estimate all parameters by treating $\boldsymbol{S}$ as missing observations and $\boldsymbol{A}$ as model parameters and iteratively update the estimation of $\boldsymbol{S}$ and $\boldsymbol{A}$, as illustrated in Section IV-B.
2) Because $\boldsymbol{\Lambda}$ is anisotropic, the resulting distance is Mahalanobis distance that can account for the noise heterogeneous effect, rather than the Euclidean distance that ignores band-dependent noise characteristics.
3) $\boldsymbol{S}$ is unknown and treated as missing observations in the EM algorithm framework. We use $E(\boldsymbol{S})$ as estimation of $\boldsymbol{S}$. To estimate parameters in $E(\boldsymbol{S})$, we use $\boldsymbol{X}$ as input to FCNN and optimize FCNN parameters. Given the estimated parameters in FCNN, we achieve $\hat{\boldsymbol{S}} = E(\boldsymbol{S})$, as illustrated in Section IV-C.
4) When estimating parameters in FCNN for obtaining $E(\boldsymbol{S})$, we use a reconstruction loss based on $(\boldsymbol{X} - \boldsymbol{A}E(\boldsymbol{S}))^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}E(\boldsymbol{S}))$, which incorporates the SMM to ensure meaningful $\boldsymbol{S}$ estimation, as illustrated in Section IV-C.
5) The second term in the objective function (24) is used to regulate the endmembers, and α defines the relative importance of this regulation. The purified means approach is adopted to estimate $\boldsymbol{A}$, as illustrated in Section IV-D.

The proposed BCUN framework is illustrated in Fig. 2.

B. EM Iteration

The EM algorithm is an iterative approach that is widely used to optimize the incomplete data problem by iterating between the estimation of model parameters, given missing observations, and the estimation of missing observations, given the model parameters [33]. In this article, we treat the abundances $\boldsymbol{S} = \{\boldsymbol{s}_i|i = 1, 2, \ldots, N\}$ as missing observations and endmembers $\boldsymbol{A}$ as model parameters to iteratively estimate both $\boldsymbol{A}$ and $\boldsymbol{S}$. The E-step is the computation of the expectation over the entire range of possible values of $\boldsymbol{S}$, i.e., $E(\boldsymbol{S})$, and the M-step updates $\boldsymbol{A}$ by minimizing (26) given $E(\boldsymbol{S})$. The main steps in the EM algorithm to estimate $\boldsymbol{S}$ and $\boldsymbol{A}$ are summarized as follows.

1) *Initialization:* Set the initial value for $\boldsymbol{A}$. The VCA algorithm [23], a fast and popular unsupervised endmember extraction approach, is used to estimate the initial value of $\boldsymbol{A}$.
2) *E-Step:* Given endmembers $\boldsymbol{A}$ and the noise variance $\boldsymbol{\Lambda}$, estimate abundances $\boldsymbol{S}$ by optimizing an FCNN, as introduced in Section IV-C.
3) *M-Step:* Given $\boldsymbol{S}$, estimate endmembers $\boldsymbol{A}$ and the noise variance $\boldsymbol{\Lambda}$. Endmembers $\boldsymbol{A}$ are estimated using purified means approach [25], which is discussed in Section IV-D. The noise variance $\boldsymbol{\Lambda}$ is estimated by the reconstructed residual, which is introduced in Section IV-E.

C. FCNN Training

The objective of the E-step of the EM algorithm introduced in Section IV-B is to obtain the estimated abundance $\hat{S}$ which is achieved by an FCNN model in (8).

To estimate $E(\boldsymbol{S})$, we first need to estimate the parameters in FCNN, i.e., $\boldsymbol{\beta}$. Here, we construct the following objective function to estimate $\boldsymbol{\beta}$:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}}\left\{\left((\boldsymbol{X} - \boldsymbol{A}f(\boldsymbol{X}, \boldsymbol{\beta}))^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}f(\boldsymbol{X}, \boldsymbol{\beta}))\right)\right\} \tag{25}$$

where $(\boldsymbol{X} - \boldsymbol{A}f(\boldsymbol{X}, \boldsymbol{\beta}))^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{X} - \boldsymbol{A}f(\boldsymbol{X}, \boldsymbol{\beta}))$ is the Mahalanobis distance to account for the noise heterogeneous effect. The Adam stochastic optimizer [34] is adopted in this work to estimate $\boldsymbol{\beta}$.

D. Purified Means

For the M-step of the iterative EM algorithm introduced in Section IV-B, the update of endmember $\boldsymbol{A}$ is achieved

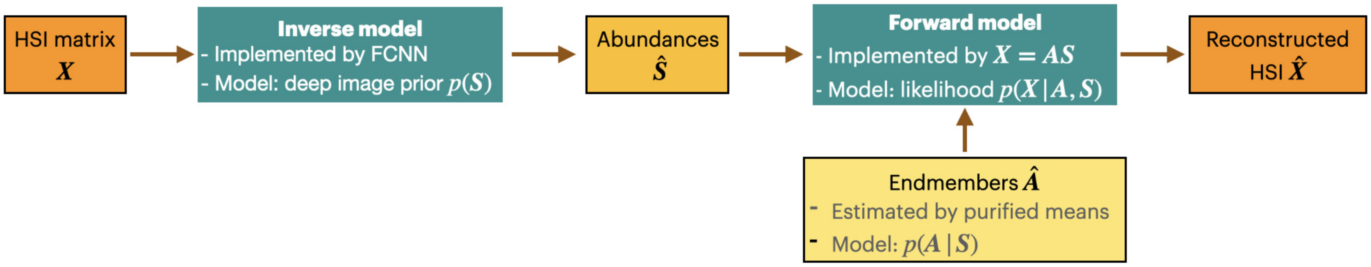

HSI matrix X
Inverse model
- Implemented by FCNN
- Model: deep image prior $p(S)$
Abundances $\hat{S}$
Forward model
- Implemented by $X = AS$
- Model: likelihood $p(X|A,S)$
Reconstructed HSI $\hat{X}$
Endmembers $\hat{A}$
- Estimated by purified means
- Model: $p(A|S)$

Fig. 2. BCUN framework. The inverse unmixing model is implemented by FCNN, where DIP $p(\boldsymbol{S})$ is used to model the spatial correlation of the abundance field. The LSMM works as the forward model to reconstruct the HSI using the estimated $\boldsymbol{A}$ and $\boldsymbol{S}$. The data likelihood $p(\boldsymbol{A}|\boldsymbol{S})$ designed based on it is integrated into the Bayesian framework. The endmember $\boldsymbol{A}$ is modeled and estimated by a "purified means" approach that can be seamlessly integrated into the Bayesian framework by a designed conditional distribution of the endmembers given the abundance $p(\boldsymbol{A}|\boldsymbol{S})$.

by minimizing the following objective function, which is the second term in (24) constrained by the first term in (24)

$$J_A = \arg\min_{\boldsymbol{a}_k} \sum_{j=1}^{K} \left|\left|\left(\boldsymbol{a}_k - E\left(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right)\right)\right|\right|^2$$

$$\text{s.t. } \boldsymbol{x}_i - \sum_{j=1}^{K} \boldsymbol{a}_j s_i^j = 0 \tag{26}$$

where $E(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k})$ is the conditional expectation of the kth endmember given $\boldsymbol{S}$ and all the other endmembers. Here, $E(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k})$ is estimated as the mean value of all purified pixels of the kth endmember $\boldsymbol{y}_i^k$, i.e., the pixels that are purified by removing the contribution of all the endmembers other than the kth endmember

$$E\left(\boldsymbol{a}_k|\boldsymbol{S}, \boldsymbol{a}_{j\neq k}\right) = \frac{1}{N}\sum_{i}^{N} \boldsymbol{y}_i^k \tag{27}$$

where N is the total number of pixels in the image. The constraint in (26) is implemented in the process of obtaining the purified pixel $\boldsymbol{y}_i^k$, which is formulated as follows [25]:

$$\boldsymbol{y}_i^k = \left(\boldsymbol{x}_i^k - \sum_{j\neq k}^{K} s_i^j \boldsymbol{a}_j\right) \bigg/ s_i^k, \quad s_i^k > 0. \tag{28}$$

E. Noise Variance $\boldsymbol{\Lambda}$ Update

Inspired by related researches [10], the noise variance $\boldsymbol{\Lambda}$ is estimated by calculating the variance of the reconstructed residual, which is formulated as follows:

$$\boldsymbol{r}_i = \boldsymbol{x}_i - \mathbf{A}\boldsymbol{s}_i \quad \text{for} \quad i = 1, 2, \ldots, N \tag{29}$$

$$\boldsymbol{\Lambda} = \text{VAR}(\{\boldsymbol{r}_i\}) \tag{30}$$

where VAR($\mathbf{t}$) is the function of calculating the variance of variable $\mathbf{t}$.

F. Summary of Complete Algorithm

Based on the EM steps described in Section IV-B, the complete algorithm used for solving BCUN can be achieved, which is summarized in Algorithm 1.

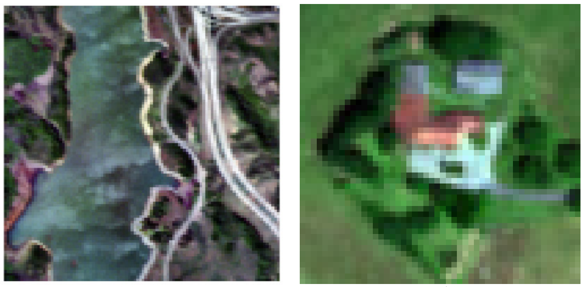
Algorithm 1 BCUN

```
Input: HSI $X$, numbers of endmembers $K$, and iteration numbers $\tau$
Output: endmember matrix $\hat{A}$, abundance matrix $\hat{S}$
Initialization: $t := 1$, $A^{(0)} = \text{VCA}(X)$, $\Lambda^{(0)} = \text{VAR}(X)$
  While $t \leqslant \tau$ do
    E-step:
        estimate $\beta$, given $\{X, A, \Lambda\}$ according to Eq. 25.
        estimate $S$ by $f(X, \beta)$ in Eq. 8
    M-step:
        for k = $1, 2, \ldots, K$
          estimate $\{y_i^k\}$ according to Eq. 28
          estimate $a_k$ using $\{y_i^k\}$ according to Eq. 27
        end for
        update $\Lambda$ according to Eq. 29 and 30
  end while
Note: $A = \{a_k|k = 1 : K\}$ and $S = \{s_i|i = 1 : N\}$
```

V. EXPERIMENTS

A. Datasets

1) Simulated HSI: In this experiment, we simulate a 104 × 104 sized HSI with four endmembers with 200 bands. Each pixel in the simulated HSI is a mixture of the four endmembers. Mixed pixels are created using the four endmembers multiplied by four abundance maps following LSMM. Abundance maps are generated by first being divided into 8 × 8 sized homogeneous blocks of one of the four endmembers and then degrading the blocks by applying a spatial low-pass filter of 9 × 9. The resulting HSI is further degraded by zero-mean Gaussian noise with different noise variances in different bands. The band-dependent SNR values used for simulation are estimated from the benchmark Indian Pines image. Suppose that the estimated SNR vector $\boldsymbol{q}$ has been centralized and normalized, and the simulated SNR vector $\boldsymbol{r}$ can be obtained following the rule of $\boldsymbol{r} = \gamma\boldsymbol{q} + \boldsymbol{c}$ [10], where γ is the amplitude that determines the magnitude of fluctuation of band-dependent SNR values and c is the center value that defines the overall SNR of all bands. Three HSIs with different noise levels (SNR = 10, 20, and 30 dB) are simulated by fixing $\gamma = 5$ and varying c.



(a) (b)

Fig. 3. RGB images of real HSIs. (a) Jasper. (b) Saint.

2) Jasper Ridge HSI Dataset: Jasper Ridge is a popular HSI with 512×614 pixels. Each pixel is recorded at 224 channels ranging from 0.38 to 2.5 μm. Due to the difficulties of ground-truth acquisition, a subimage with 100×100 was selected, and its true endmember and abundance were collected. After removing channels 1–3, 108–112, 154–166, and 220–224 (due to dense water vapor and atmospheric effects), 198 channels remained. There are four endmembers in this data, i.e., "road," "soil," "water," and "tree." The ground truth for the dataset is generated by Zhu [35], which has been widely used [36]–[38]. The red, green, and blue (RGB) color composition of the HSI is shown in Fig. 3(a).

3) Saint Andre HSI Dataset: The Saint Andre HSI dataset is used in an unmixing paper [39] and published on the website https://zenodo.org/record/2142185#.YWbDyC0RoUt. The HSI contains 50×50 pixels, composed of 415 spectral bands ranging from VNIR to SWIR (0.40–2.40 μm). The spectral bands with strong noise in the spectral ranges 1.34–1.55 and 1.80–1.98 μm have been removed. The RGB color composition of the HSI is shown in Fig. 3(b). Endmember spectra of six distinct materials (i.e., "tree," "grass," "soil," "road," "building 1," and "building 2") have been manually extracted based on prior knowledge of the scene. This HSI dataset has only true endmembers without true abundances. Methods are evaluated mainly based on their performances on the endmember extraction. Abundances are also evaluated visually by comparing them with the RGB image.

B. Experimental Setup

1) Methods Compared: The proposed method BCUN is compared with several traditional endmember extraction method, including fast autonomous spectral endmember determination (N-FINDR) [40], PPI [24], VCA [23], K-P-Means [25], and several state-of-the-art methods, including the spatial group sparsity regularized nonnegative matrix factorization (SGSNMF) approach [41], untied denoising autoencoder with sparsity (uDAs) [13], [42], and a typical linear plug-and-play (PnP) priors framework, PnP [43]–[45], which are proposed recently. PnP and uDAs are deep learning-based methods. It should be noted that K-P-Means, uDAs, and PnP provide estimations of endmembers and abundances, simultaneously, while, for N-FINDR, PPI, and VCA, only endmembers are estimated, and the abundances are obtained in a postprocessing stage by using the Sunsal method. K-P-Means is also solved by the EM algorithm where the E-step estimates abundances using Sunsal and M-step achieves endmembers update via purified means. Therefore, by comparing the performances between K-P-Means, the advantage of the proposed method in terms of further improvement over K-P-Means can be demonstrated. Moreover, since uDAs, PnP, and the proposed method are deep neural network-based approaches but with vastly different implementations, the comparison between the two can justify the proposed BCUN approach and other contributions in the context of the proposed Bayesian MAP optimization framework. To demonstrate the advantage of addressing the heterogeneous noise effect, we design the BCUN0 method by replacing the Mahalanobis-distance loss in BCUN with the Euclidean-distance loss and compare it with BCUN.

2) Numerical Measures: The spectral angle distance (SAD) defined as SAD $= \cos^{-1}((\boldsymbol{a}^T\widehat{\boldsymbol{a}})/(\|\widehat{\boldsymbol{a}}\|\|\boldsymbol{a}\|))$ and the spectral information divergence (SID) defined as SID $= D(\boldsymbol{a}/\widehat{\boldsymbol{a}}) + D(\widehat{\boldsymbol{a}}/\boldsymbol{a})$ are used to measure the precision of the endmember extraction, where $D(\boldsymbol{x}/\boldsymbol{y})$ measures the relative entropy between $\boldsymbol{x}$ and $\boldsymbol{y}$. The accuracy of abundance estimation is measured using the abundance angle distance (AAD) and the abundance information divergence (AID) by replacing $\boldsymbol{a}$ with $\boldsymbol{s}$ in the above SAD and SID equations, as well as the MSE.

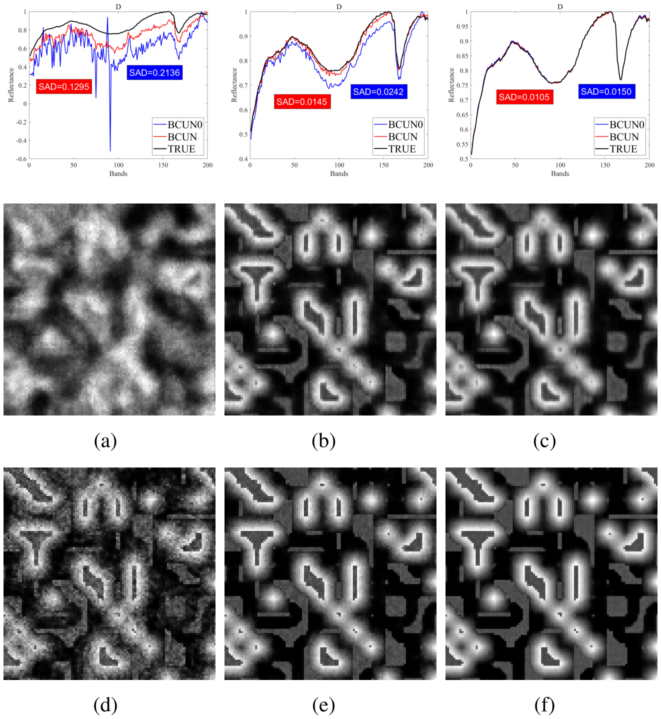
3) Implement Settings: For the test on the simulated HSI, the learning rate of BCUN is empirically set as 0.00025. The number of EM iterations is 50. For each EM iteration, we use 20 epochs to train the neural network. We adopt three skip connections in the FCNN, as shown in Fig. 1. For the test on the real Jasper HSI, the learning rate of BCUN is set as 0.03. The number of EM iterations is 60. For each EM iteration, the network is trained 200 times. We adopt one skip connection in the FCNN for the Jasper HSI dataset. For the test on the real Saint HSI, the learning rate of BCUN is set as 0.001. The number of EM iterations is 10. For each EM iteration, the network is trained 150 times. We adopt two skip connections in the FCNN for the Saint HSI dataset.

Model parameters, including the number of layers, the learning rate, and iteration times, are determined empirically. Hyperparameters are required to adjust for different HSIs since HSIs vary in the sensor, data size, and complexity. The real HSIs contain fewer mixed pixels and spatial texture features, which can be unmixed with a simpler network. Thus, we reduce the layers of the network to accelerate the model training. The learning rate needs to be adjusted to suit the network architecture. All data processing is conducted using the Python language under the Pytorch framework.

The benchmark PnP, uDAs, and SGSNMF methods were implemented in MATLAB 2018 under an Intel Xeon Silver 4110 CPU @ 2.10 GHz. The VCA, PPI, N-FINDR, KPmeans, and proposed BCUN methods were implemented in the Pytorch toolbox with an NVIDIA GeForce RTX 2080 Ti GPU.

C. Simulated Study

To compare the performances between traditional MSE loss that is built upon the Euclidean distance in BCUN0 and the proposed Mahalanobis distance loss in BCUN, we first apply BCUN and BCUN0 on the three simulated HSIs with SNR = 10, 20, and 30 dB. Fig. 4 shows the endmember (the first row)



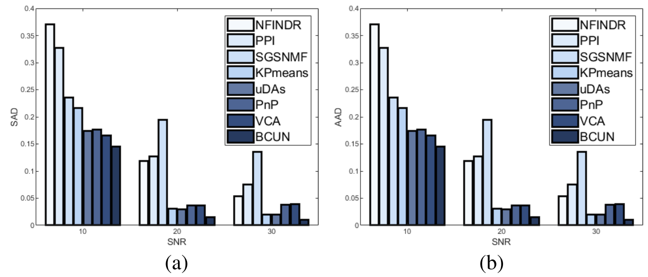
(a) (b) (c)

(d) (e) (f)

Fig. 4. Endmember (the first row) and abundance maps of one endmember achieved by of BCUN and BCUN0 (the second and the third row separately) with different SNR values, i.e., 10, 20, and 30 dB, respectively, from left column to right column. (a) AAD = 0.5678. (b) AAD = 0.1732. (c) AAD = 0.1636. (d) AAD = 0.4573. (e) AAD = 0.1023. (f) AAD = 0.0735.

and abundance maps of one endmember achieved by BCUN and BCUN0 (the second and third rows separately) with different SNR values, i.e., 10, 20, and 30 dB. On all endmember figures (in the first row), the endmember extracted by BCUN (red lines) is closer to the true endmember (black lines) than the endmember estimated by BCUN0 (blue lines). In addition, endmembers achieved by BCUN are smoother than those obtained by BCUN0, which is specifically obvious when the noise level is high [e.g., in Fig. 4 (left) where SNR = 10 dB]. This demonstrates that the proposed M-distance loss in FCNN can better deal with the heterogeneous noise than the traditional MSE loss for enhanced endmember and abundance estimation, especially when the noise level of HSI is high. Fig. 4 shows negative values in endmembers at SNR = 10 because BCUN0 does not consider the noise heterogeneous effect and leads to a large bias in endmember estimation at some spectral bands. The endmembers estimated by BCUN are all positive at any noise level. It also shows the importance of addressing noise heterogeneity, especially when SNR values are low.

To evaluate the performance of the proposed BCUN, we compare it with the other methods introduced in Section V-B1 by testing all methods with ten independent runs on three simulated HSIs with different SNR levels. The mean values and the standard deviation (std) values of SAD, AAD, SID, AID, and MSE are summarized in Table I. The running times of different SU methods are summarized in Table II. The methods' comparisons in terms of SAD and AAD are also illustrated in Fig. 5. Given that the smallest SAD and AAD mean values were all achieved by BCUN, BCUN outperformed all the other methods on endmember extraction and abundance estimation at all noise levels, indicating that the proposed BCUN is able to more accurately model the abundance and endmember information in mixed pixels by accounting for the heterogeneous noise and the spatial correlation effect in a Bayesian FCNN framework. In particular, the observation that the mean AAD values achieved by BCUN are smaller than those achieved by the other values clearly reflects the advantages of the designed BCUN approach over the traditional Sunsal approach for abundance mapping. Some abrupt changes and spatial detail information were well delineated by BCUN, which justified the use of DIP to capture the nonstationary spatial correlation information in HSI. Convolutional layers tend to smooth features, but the FCNN structure is able to capture the spatial textural information, which is DIP. The methods' comparison in terms of SAD and AAD is also illustrated in Fig. 5.



(a) (b)

Fig. 5. AAD and AAD bar graphs achieved by different methods with different SNR values, i.e., 10, 20, and 30 dB. (a) SAD. (b) AAD.

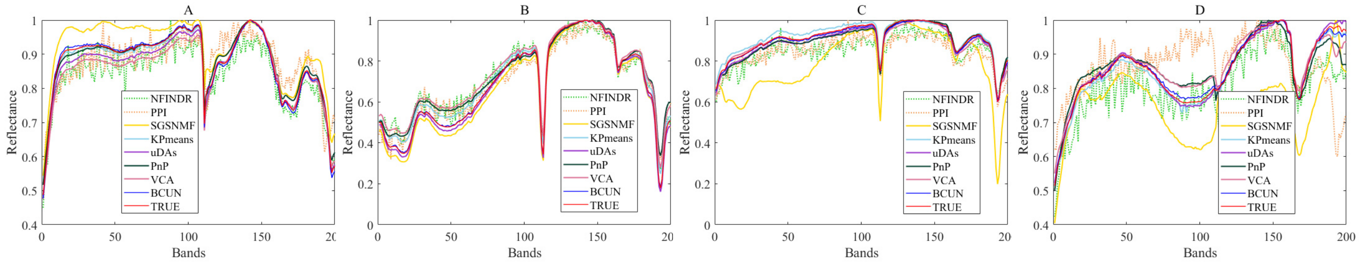
Fig. 6 shows the four endmembers estimated by different SU methods and the true endmember (the red line) at the noise level of SNR = 30 dB. The proposed method BCUN appears to achieve the closest endmember spectrum (dark blue lines) to the true endmember (red lines) for all four endmembers. Endmembers estimated by deep learning-based uDAs and PnP algorithms (purple and dark green lines) are closer to the true endmembers than the other traditional methods but also introduce an obvious bias away from the true endmember 1. Endmembers achieved by K-P-Means (light blue lines) are smooth and very close to the true endmembers 1 and 4 but have a larger bias away from true endmembers 2 and 3. VCA (pink lines) generates endmembers relatively smooth with a very large bias from the true endmembers. Endmembers obtained by PPI and N-FINDR (orange and green dotted lines) are very noisy, indicating that these two methods are sensitive to resist noise. These visual interpretations align with the quantitative results in Table I.

Fig. 7 displays the abundance maps of one endmember generated by PPI, N-FINDER, VCA, K-P-Means, uDAs, PnP, and the proposed BCUN. As we can see, BCUN achieved abundance maps that are very close to the ground-truth maps. K-P-Means tends to preserve some noise due to the failure to account for the spatial correlation effect in HSI, which is especially true when the noise level is high (i.e., SNR = 10 and 20 dB) where the abundance maps of these traditional

TABLE I

AVERAGE SAD, AAD, SID, AID, AND MSE OF ABUNDANCES OBTAINED FROM TEN INDEPENDENT RUNS BY DIFFERENT METHODS USING THE SIMULATED DATA OVER SNR FROM 10 TO 30 dB. THE BEST RESULTS ARE IN BOLD

SNR=10, 4 endmembers, 10816 pixels					
Method	SAD	AAD	SID	AID	MSE
NFINDR	0.3704±3.09%	0.7009±2.38%	0.1537±0.0308%	6.0106±0.3139%	0.0469
PPI	0.3271±5.56%	0.7284±**1.47%**	0.1791±0.0071%	6.3262±**0.1043%**	0.0559
SGSNMF	0.2360±7.51%	0.5704±2.37%	0.3374±0.0869%	4.6836±0.2111%	0.0351
KPmeans	0.2162±4.93%	0.7208±4.50%	0.0268±0.0183%	6.5955±0.6952%	0.0475
uDAs	0.1736±2.67%	0.7043±6.17%	0.0185±0.0111%	6.3495±1.2081%	0.0432
PnP	0.1766±**1.98%**	0.7422±7.66%	0.0201±**0.0063%**	7.8469±2.1994%	0.0580
VCA	0.1659±3.37%	0.7745±7.55%	**0.0156**±0.0098%	8.9710±2.1885%	0.0560
BCUN	**0.1449**±9.28%	**0.4573**±12.68%	0.0394±0.0545%	**3.7158**±0.8387%	**0.0161**
SNR=20, 4 endmembers, 10816 pixels					
Method	SAD	AAD	SID	AID	MSE
N-FINDR	0.1184±**0.21%**	0.4280±**1.03%**	0.0168±0.0011%	3.4733±0.1174%	0.0217
PPI	0.1330±1.97%	0.6112±7.20%	0.0210±0.0037%	5.1706±0.3289%	0.0354
SGSNMF	0.1950±13.55%	0.4832±17.36%	0.1609±0.1651%	4.0203±2.0509%	0.0154
KPmeans	0.0308±0.90%	0.2979±4.92%	0.0012±0.0008%	1.9187±0.5029%	0.0100
uDAs	0.0294±1.51%	0.2451±4.07%	0.0014±0.0014%	1.4097±0.4577%	0.0060
PnP	0.0364±0.96%	0.2788±6.98%	0.0020±0.0008%	1.6709±1.0786%	0.0064
VCA	0.0362±1.21%	0.3055±6.18%	0.0020±0.0012%	1.8566±0.7442%	0.0081
BCUN	**0.0145**±0.59%	**0.1023**±1.70%	**0.0004±0.0002%**	**0.5288±0.1152%**	**0.0012**
SNR=30, 4 endmembers, 10816 pixels					
Method	SAD	AAD	SID	AID	MSE
N-FINDR	0.0537±0.75%	0.2215±5.88%	0.0038±0.0009%	1.3271±0.5318%	0.0039
PPI	0.0752±1.35%	0.5262±2.98%	0.0235±**0.0000%**	32.1737±**0.0006%**	0.0334
SGSNMF	0.1359±13.19%	0.3427±9.88%	0.1660±0.3383%	2.2419±0.9875%	0.0201
KPmeans	0.0195±**0.48%**	0.1126±2.68%	0.0010±0.0003%	0.4808±0.2388%	0.0014
uDAs	0.0195±1.10%	0.1377±4.95%	0.0007±0.0010%	0.6136±0.3938%	0.0034
PnP	0.0383±0.81%	0.2657±5.12%	0.0025±0.0007%	1.4322±0.5323%	0.0060
VCA	0.0396±1.03%	0.2321±7.14%	0.0026±0.0009%	1.0945±0.6274%	0.0041
BCUN	**0.0105**±0.56%	**0.0735±1.14%**	**0.0002**±0.0002%	**0.2965**±0.1047%	**0.0010**



(a) (b) (c) (d)

Fig. 6. Endmembers achieved by different methods when SNR equals 30 dB. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4.

TABLE II

RUNNING TIMES OF DIFFERENT METHODS ON THE SIMULATED DATA WITH SNR = 30 dB

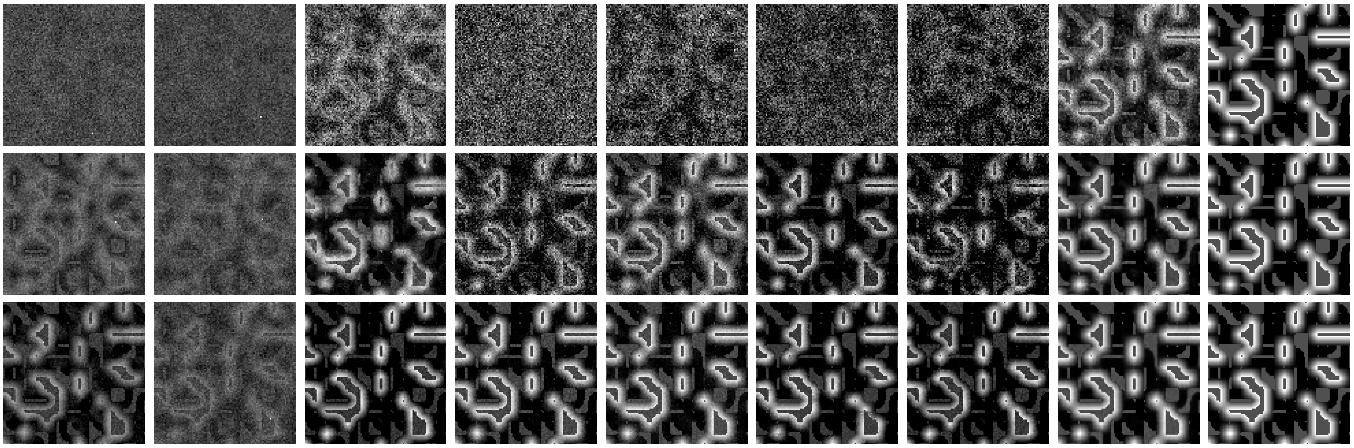
Methods	**VCA**	**PPI**	**BCUN**	**N-FINDR**	**KPmeans**	**PnP**	**uDAs**	**SGSNMF**
Time (s)	**0.0778**	0.1641	18.6107	0.3563	43.7626	12.1875	24.5168	91.0099

methods are very noisy. The state-of-the-art methods, uDAs and PnP, achieve smoother and clearer abundance maps than other methods but are still noisy than BCUN at all noise levels.

D. Test on Real HSIs

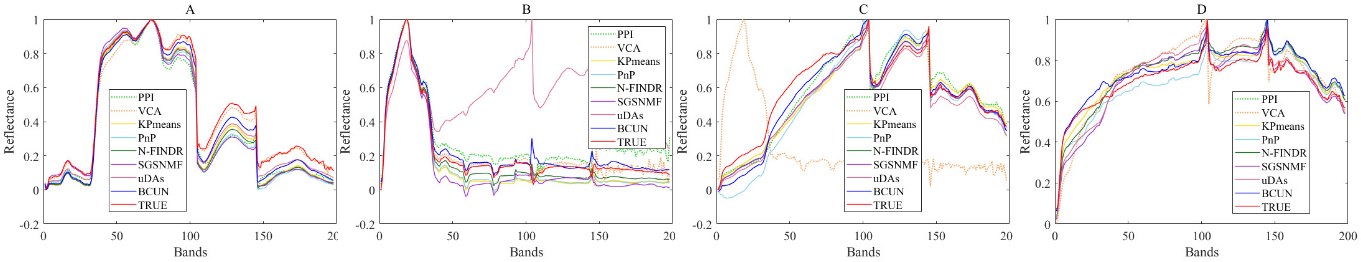
1) Jasper Ridge Scene: All methods are conducted on the Jasper Ridge HSI. The estimated endmembers and abundances are compared in Figs. 8 and 9 separately. Numerical measurements achieved by all methods are summarized in Table III. Table III indicates that BCUN achieves the lowest mean values of SAD, SID, AAD, and AID among all methods tested, indicating that BCUN extracts endmembers and abundances more accurately than other methods.

Fig. 9 shows that overall BCUN is more capable of generating abundance maps that are close to the ground-truth maps in terms of both the intense brightness and the structural



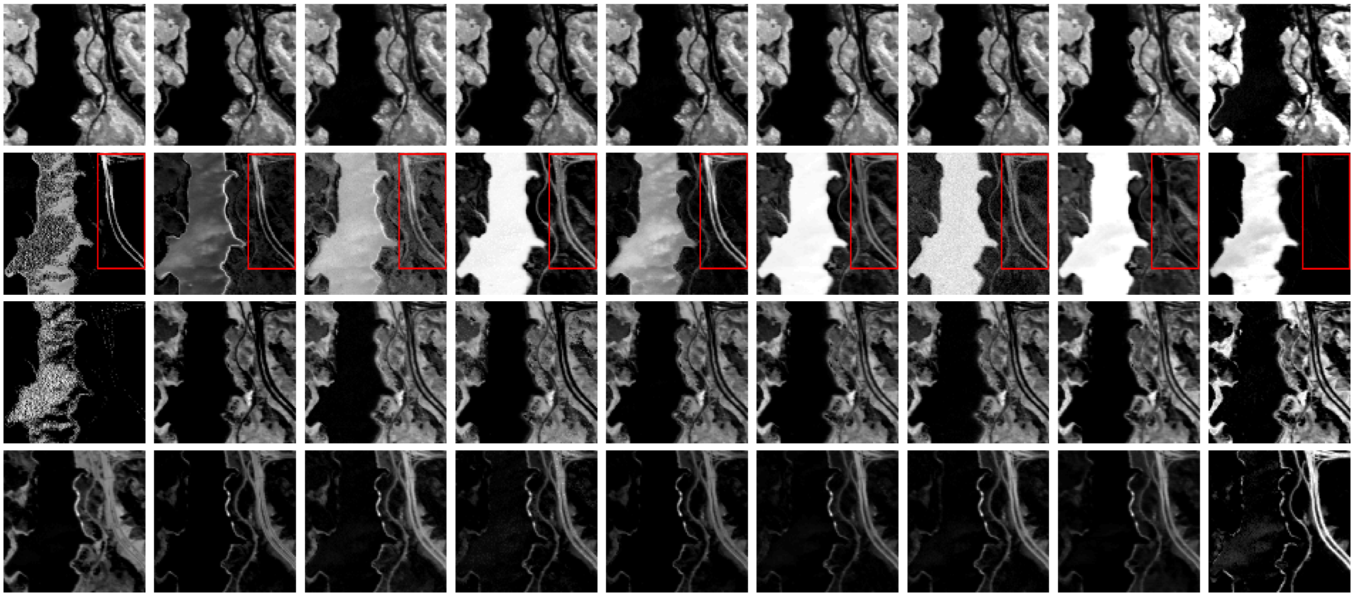
(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 7. Abundance maps achieved by different methods on one endmember with different SNR values, i.e., 10, 20, and 30 dB, from the top row to bottom row. Brighter pixels indicate high abundance while darker pixels indicate low abundance. (a) N-FINDR. (b) PPI. (c) SGSNMF. (d) KPmeans. (e) uDAs. (f) PnP. (g) VCA. (h) BCUN. (i) Ground truth.



(a) (b) (c) (d)

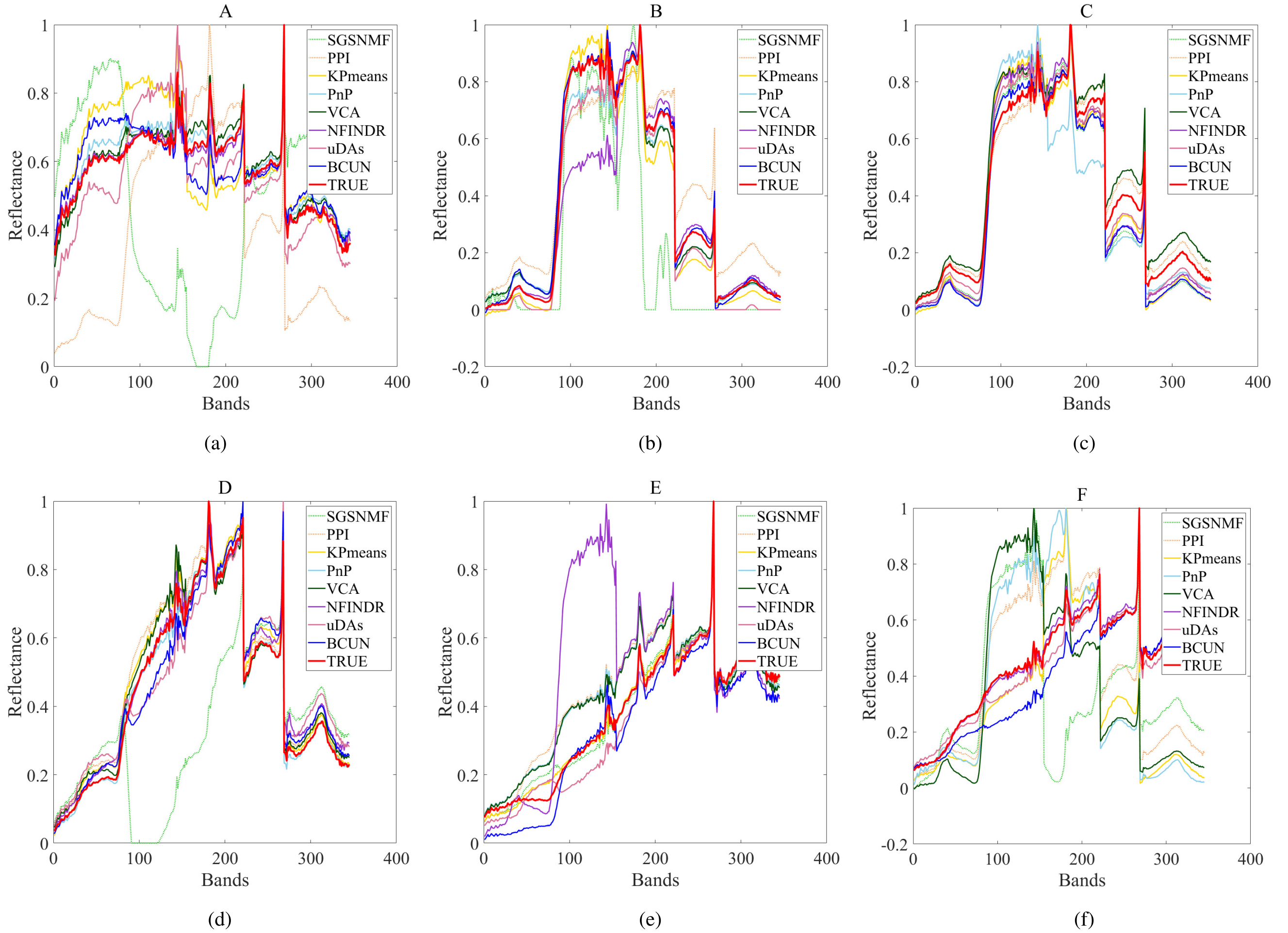
Fig. 8. Estimated endmembers achieved by different methods for Jasper Ridge HSI data along with the true endmember. (a) Tree. (b) Water. (c) Soil. (d) Road.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 9. Abundance maps achieved by different methods on four endmembers (tree, water, soil, and road), respectively, from the top row to bottom row for Jasper Ridge HSI data. Brighter pixels indicate high abundance, while darker pixels indicate low abundance. It shows that overall BCUN is more capable of generating abundance maps that are close to the ground-truth maps in terms of both the intense brightness and the structural characteristics. For example, the bright water area in the second row generated by BCUN is closer to the intensity brightness of the ground truth, and also BCUN generates the highlighted red box area that is closest to the ground truth, whereas the red box areas of the other methods wrongly highlight the road as water in the water abundance maps. (a) PPI. (b) VCA. (c) KPmeans. (d) PnP. (e) N-FINDR. (f) SGSNMF. (g) uDAs. (h) BCUN. (i) Ground truth.

characteristics. For example, the bright water area in the second row of Fig. 9 generated by BCUN is closer to the intensity brightness of the ground truth, and also BCUN generates the highlighted red box area in Fig. 9 that is closest



(a) (b) (c)

(d) (e) (f)

Fig. 10. Estimated endmembers achieved by different methods for Saint HSI data along with the true endmember. Endmembers extracted by the BCUN method (blue line) are the closest to the true endmembers (red line) over six categories. Although some methods perform very well on specific endmembers (e.g., uDAs on soil and N-FINDR on building 2), they generate big bias on some other endmembers (e.g., uDAs on road and N-FINDR on building 1). BCUN is demonstrated to be able to distinguish and extract different endmembers well. (a) Road. (b) Tree. (c) Grass. (d) Soil. (e) Building1. (f) Building2.

to the ground truth, whereas the red box areas of the other methods wrongly highlight the road as water in the water abundance maps.

Fig. 8 shows the four endmembers achieved by different methods and the true endmembers (red line). On average, endmembers obtained by the proposed BCUN method are the closest to the true endmembers among all methods, which is consistent with the SAD statistics in Table III.

2) Saint Andre Scene: Table IV demonstrates that the proposed BCUN achieves the best endmember estimation performance with the lowest SAD and SID values compared with other methods. Fig. 10 displays six endmembers extracted by different methods. Endmembers extracted by the BCUN method (blue line) are the closest to the true endmembers (red line) over six categories. Although some methods perform very well on specific endmembers (e.g., uDAs on soil and N-FINDR on building 2), they generate big bias on some other endmembers (e.g., uDAs on road and N-FINDR on building 1). BCUN is demonstrated to be able to distinguish and extract different endmembers well.

TABLE III

AVERAGE SAD, SID, AAD, AID, AND MSE OF ABUNDANCES OBTAINED BY DIFFERENT METHODS FOR JASPER RIDGE HSI DATA. THE BEST RESULTS ARE IN BOLD

Jasper Ridge HSI, 4 endmembers, 10000 pixels					
Methods	SAD	SID	AAD	AID	MSE
PPI	0.3085	0.3710	0.6809	8.5008	0.0898
VCA	0.2537	0.1589	0.4288	4.1735	0.0530
KPmeans	0.1680	0.1190	0.3938	4.1484	0.0555
PnP	0.1622	0.1322	0.3116	2.6532	0.0202
N-FINDR	0.1604	0.0565	0.3537	3.0730	0.0278
SGSNMF	0.1392	0.1571	0.2897	2.8536	**0.0201**
uDAs	0.1181	0.0624	0.3044	2.8851	0.0242
BCUN	**0.1065**	**0.0383**	**0.2892**	**2.4576**	0.0207

Since the true abundances were not provided in the HSI dataset, we visually evaluate abundances by comparing abundance maps with the RGB image. Abundances of road and tree are presented in Figs. 11 and 12 separately. BCUN extracts the tree endmember very well by achieving the endmember spectra (blue line) very close to the true endmember (red line) [see Fig. 10(b)]. Correspondingly, the tree abundance map generated by BCUN [see Fig. 11(h)] shows very clear tree positions and edges, which is highly consistent with the RGB

TABLE IV

AVERAGE SAD AND SID OBTAINED BY DIFFERENT METHODS FOR SAINT HSI DATA. THE BEST RESULTS ARE IN BOLD. THIS HSI PROVIDE ONLY TRUE ENDMEMBERS MANUALLY EXTRACTED BASED ON PRIOR KNOWLEDGE OF THE SCENE WITHOUT TRUE ABUNDANCES

Saint HSI, 6 endmembers, 2500 pixels		
Methods	SAD	SID
SGSNMF	0.4462	2.8081
PPI	0.2327	0.1614
KPmeans	0.1992	0.2286
PnP	0.1930	0.1872
VCA	0.1813	0.1732
N-FINDR	0.1343	0.0503
uDAs	0.1292	0.4733
BCUN	**0.1137**	**0.0408**

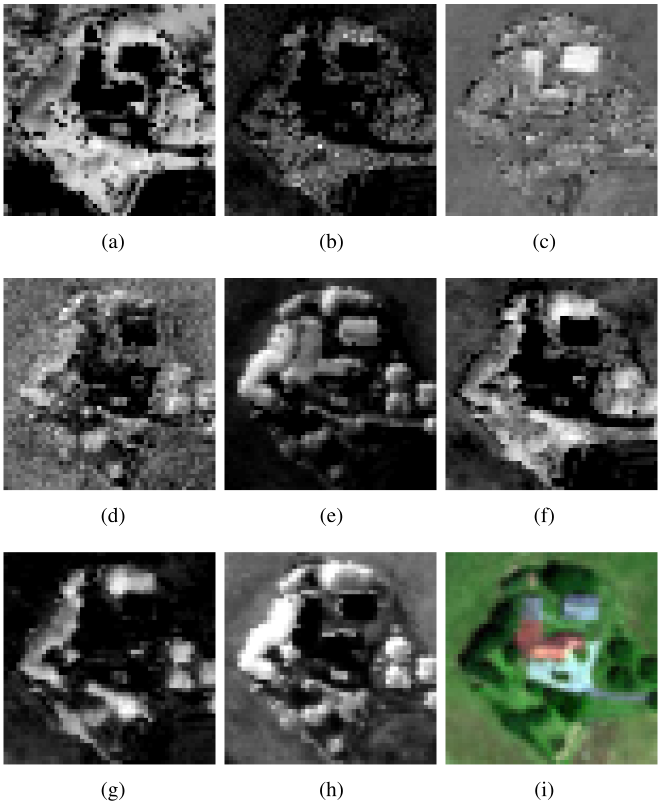


Fig. 11. Abundance maps of tree achieved by different methods on Saint HSI data. Brighter pixels indicate high abundance, while darker pixels indicate low abundance. BCUN extracts the tree endmember very well [see Fig. 10(b)]. Correspondingly, the tree abundance map generated by BCUN (h) shows very clear tree positions and edges, which is highly consistent with the RGB image. PPI, N-FINDR, and uDAs cannot distinguish tree and grass very well. PnP, uDAs, and VCA tend to preserve more noise than BCUN. KPmeans shows less spatial texture information than BCUN. (a) PPI. (b) PnP. (c) N-FINDR. (d) uDAs. (e) SGSNMF. (f) VCA. (g) KPmeans. (h) BCUN. (i) RGB image.

image. Some of the other methods (e.g., PPI, N-FINDR, and uDAs) cannot distinguish tree and grass very well. PnP, uDAs, and VCA tend to preserve more noise than BCUN. KPmeans shows less spatial texture information than BCUN.

BCUN outperforms other methods in spatial information preservation by delineating the road outline clearly and smoothly. Although the road abundance achieved by BCUN [see Fig. 12(g)] preserves some other objects in the background, the road object is still the brightest area, which is easily distinguished from other objects. It is reasonable that the abundance of buildings is not zero on the road abundance map because building and road may contain similar physical materials.

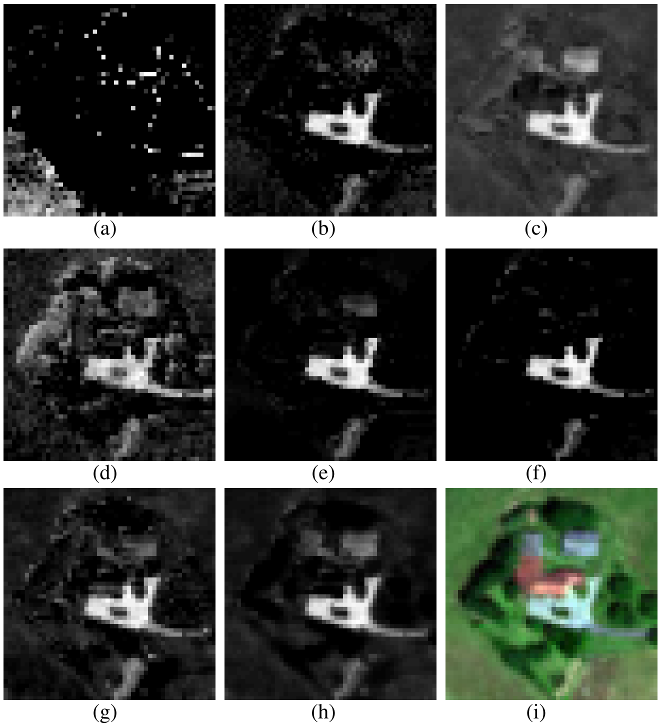


Fig. 12. Abundance maps of road achieved by different methods on Saint HSI data. Brighter pixels indicate high abundance, while darker pixels indicate low abundance. BCUN outperforms other methods in spatial information preservation by delineating the road outline clearly and smoothly. (a) PPI. (b) PnP. (c) N-FINDR. (d) uDAs. (e) SGSNMF. (f) VCA. (g) KPmeans. (h) BCUN. (i) RGB image.

VI. CONCLUSION

In this article, we presented a Bayesian framework for hyperspectral unmixing in which the EM algorithm was applied to solve the MAP problem. Unlike most deep learning-based SU techniques, BCUN adopted FCNN rather than fully connected layers or Sunsal to better exploit the spatial correlation effect in HSI for enhanced abundance modeling and estimation. In addition, the noise heterogeneity effect in HSI was addressed by the M-distance loss. Conditional distribution of the endmember is designed, leading to an efficient purified means approach for endmember estimation. The above key components are seamlessly integrated into a Bayesian MAP framework, which is solved by the proposed EM approach. Therefore, the proposed BCUN approach constitutes a complete Bayesian approach with advanced modeling and optimization approaches for enhanced SU. The proposed approach was tested on both real and simulated HSIs, in comparison with several other popular SU methods, and results demonstrated that the proposed BCUN method was more capable of accurately and efficiently estimating both the endmember and abundance in HSIs.

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